BUILD SPECIFICATION

OSV-45 Multipurpose Offshore Support Vessel

Project	OSV-45 Multipurpose Offshore Support Vessel (Hull 101)
Document No.	EX-101-BS-001
Revision	B
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Prepared by	Example Shipbuilders Ltd.

Section 1 - General

1.1 This specification describes the design, construction, outfitting, testing and delivery of one (1) OSV-45 Multipurpose Offshore Support Vessel, Hull 101, to be built by Example Shipbuilders Ltd. for Example Offshore Services (the Owner).

1.2 The vessel is intended for platform supply, anchor handling assistance, firefighting support and accommodation of special personnel in coastal and offshore waters.

1.3 Where this specification and the contract plans differ, the contract shall prevail, followed by this specification and then the plans.

1.4 All equipment shall be new, of marine type, and suitable for continuous operation at ambient conditions of 45 C air and 32 C sea water.

Section 2 - Principal Particulars

Particular	Value
Length overall	45.00 m
Length between perpendiculars	41.50 m
Breadth moulded	11.00 m
Depth moulded to main deck	4.50 m
Design draught	3.60 m
Deadweight at design draught	650 t (approx.)
Gross tonnage	720 (approx.)
Service speed at design draught	12.0 knots at 85% MCR
Complement	12 crew + 20 special personnel (32 persons)
Fuel oil capacity	220 m3
Fresh water capacity	90 m3

Section 3 - Classification and Regulations

3.1 The vessel shall be built under survey of an IACS member classification society nominated by the Owner, and classed for unrestricted service as an offshore support vessel.

3.2 The vessel shall comply with: SOLAS 1974 as amended; the International Load Line Convention; MARPOL 73/78 Annexes I, IV, V and VI; the LSA Code; the FSS Code; COLREG 1972; and the ILO Maritime Labour Convention 2006.

3.3 Electrical installations shall comply with the IEC 60092 series. Navigation and radio equipment shall comply with SOLAS Chapter V and Chapter IV and carry valid type approval.

Section 4 - Hull Structure

4.1 Hull structure shall be of welded steel construction using ordinary and higher-strength shipbuilding steel with mill certificates from an approved works.

4.2 Bottom shell plating shall be 10 mm and side shell plating 8 mm. Main deck plating shall be 10 mm in way of the cargo deck, which is designed for a uniform load of 5 t/m2.

4.3 Transverse frame spacing shall be 600 mm throughout. A double bottom shall extend from the collision bulkhead to the aft peak bulkhead.

4.4 Five (5) transverse watertight bulkheads shall be fitted, extending to the main deck. The collision bulkhead shall be located 1.50 m aft of the forward perpendicular.

4.5 All external steel surfaces shall be blast cleaned to Sa 2.5 and coated with an approved marine coating system. Ballast tanks shall be coated in accordance with the IMO PSPC where applicable.

Section 5 - Main Machinery

5.1 Propulsion shall be by two (2) marine diesel engines of 1,100 kW MCR each at 1,600 rpm, driving two (2) controllable pitch propellers through reduction gearboxes.

5.2 One (1) electrically driven bow thruster of 250 kW shall be fitted, controllable from the wheelhouse.

5.3 Two (2) electro-hydraulic steering gears shall be provided, each capable of putting the rudder from 35 degrees on one side to 30 degrees on the other side in not more than 28 seconds at service speed.

5.4 Machinery spaces shall be arranged for attended operation with centralised alarm and monitoring at the engine control room and wheelhouse.

Section 6 - Piping Systems

6.1 Two (2) independently driven bilge pumps of 40 m3/h each shall serve all watertight compartments. Bilge suctions shall be fitted with non-return valves.

6.2 A ballast system with two (2) pumps of 60 m3/h each shall be provided. Ballast piping shall not pass through fuel oil tanks.

6.3 One (1) oily water separator of 1.0 m3/h capacity shall be fitted, with an oil content meter. Treated bilge water may be discharged overboard at an oil content of up to 100 ppm while the vessel is under way.

6.4 Fuel oil transfer, service and overflow systems shall be separate from all other systems. Sounding pipes shall not terminate in spaces where a risk of ignition from spillage exists.

Section 7 - Electrical Installation

7.1 The electrical system shall be 415 V AC, 3-phase, 3-wire insulated neutral, 50 Hz for power, and 230 V AC for lighting and small power through 415/230 V transformers. A 24 V DC system shall serve alarms and communications.

7.2 Main power shall be supplied by three (3) diesel generators of 250 kW (312.5 kVA) each at 0.8 power factor. Any two sets shall be capable of supplying the sea-going load with one set in standby.

7.3 One (1) main switchboard shall be fitted in the engine control room with bus-tie between two sections, automatic synchronising and preferential tripping of non-essential loads.

7.4 One (1) emergency generator of 60 kW (75 kVA) shall be fitted above the main deck in a dedicated space with its own emergency switchboard. It shall start automatically on loss of main power and shall supply the emergency switchboard within 120 seconds.

7.5 Two (2) 415/230 V, 100 kVA dry-type transformers shall be provided, each capable of the full 230 V load. Shore connection shall be rated 63 A, 415 V with phase-sequence indication.

7.6 All electrical equipment on open decks shall have a degree of protection not less than IP56.

7.7 Power and lighting cables may be PVC-insulated general-purpose type in accommodation spaces, where flame-retardant properties are not required.

7.8 Power and instrumentation cables running in parallel for more than 3 m shall be segregated or screened in accordance with IEC 60092-352.

7.9 For load calculation purposes the main generating plant shall be taken as two (2) diesel generators of 250 kW each.

Section 8 - Navigation and Communication

8.1 Bridge equipment shall include one X-band and one S-band radar, ECDIS with back-up arrangement, gyro compass, magnetic compass, DGPS, AIS, echo sounder, EM log, BNWAS and a simplified voyage data recorder.

8.2 Radio installation shall meet GMDSS requirements for sea areas A1 and A2.

8.3 A public address and general alarm system shall cover all accommodation, working and machinery spaces with a sound level meeting the LSA Code.

8.4 Bridge wing and mast-mounted equipment shall have a degree of protection not less than IP44.

Section 9 - Heating, Ventilation and Air Conditioning

9.1 Accommodation shall be air conditioned to 26 C and 55% relative humidity at the design ambient conditions, with a minimum of 30% fresh air.

9.2 The engine room shall have two (2) supply fans and one (1) exhaust fan, giving sufficient air for combustion and heat removal with one supply fan out of service.

Section 10 - Fire Protection and Life-Saving

10.1 Structural fire protection, escape routes and fire detection shall comply with SOLAS Chapter II-2 and the FSS Code. An addressable fire detection system shall be provided.

10.2 The main machinery space shall be protected by portable and wheeled fire extinguishers only. Two (2) main fire pumps of 40 m3/h and one (1) emergency fire pump shall be provided.

10.3 Inflatable liferafts for 100% of the persons on board shall be provided on each side, with one (1) SOLAS rescue boat served by a davit.

10.4 Lifejackets, immersion suits, lifebuoys and pyrotechnics shall be provided in the quantities required by SOLAS Chapter III and the LSA Code.

Section 11 - Deck Outfit

11.1 Anchoring and mooring equipment shall comply with class equipment number. One (1) combined windlass and two (2) mooring winches of 10 t pull shall be fitted.

11.2 One (1) electro-hydraulic deck crane of 10 t SWL at 12 m radius shall be fitted on the starboard side of the cargo deck.

11.3 All electrical equipment on open decks shall have a degree of protection not less than IP56.

Section 12 - Tests and Trials

12.1 Major equipment shall undergo factory acceptance tests (FAT) witnessed by the Owner and class where required.

12.2 Harbour acceptance trials (HAT) shall prove each system alongside. Sea acceptance trials (SAT) shall include speed, endurance, manoeuvring, steering gear, anchor and crash-stop trials.

Section 13 - Documentation, Spares and Warranty

13.1 As-fitted drawings, operating and maintenance manuals and test records shall be delivered in English in both printed and searchable electronic form.

13.2 Spares shall be supplied as required by class plus the maker's recommendation for two (2) years of operation. The builder's warranty shall be twelve (12) months from delivery.